

Patrick Sherlock

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Education

2019 - 2023 University of Southern California
B.S. Neuroscience, Minor Linguistics

Research Positions

2025 - present Technical Research Associate, Massachusetts Institute of Technology
Advisor: Dr. Evelina Fedorenko

2023 - 2025 Project Coordinator, Vanderbilt University
Advisor: Dr. Sarah Brown-Schmidt

2022 - 2023 Undergraduate Research Assistant, University of Southern California
Advisor: Dr. Zuzanna Fuchs

2020 - 2023 Undergraduate Research Assistant, University of Southern California
Advisors: Dr. Kirsten Lynch, Dr. Jeiran Choupan

Publications

2026 Lord, K., Kekes-Szabo, S., **Sherlock, P.**, Duff, M.C., & Brown-Schmidt, S. Moderate-severe traumatic brain injury disrupts core mechanisms of online language processing and use. *Cortex*, 197, 13-33.
DOI: 10.1016/j.cortex.2026.01.010

2026 Casto, C., Poliak, M., Tuckute, G., Small, H., **Sherlock, P.**, Wolna, A., Lipkin, B., D'Mello, A. M., & Fedorenko, E. The cerebellar components of the human language network. *Neuron*, 114, 8, 1504-1523.
DOI: 10.1016/j.neuron.2025.12.030

2023 Lan, H., Lynch, K., Custer, R., Shih, N., **Sherlock, P.**, Sepehrband, F., Toga, A., & Choupan, J. Weakly supervised perivascular spaces segmentation with salient guidance of Frangi filter. *Magnetic Resonance in Medicine*, 89, 6, 2419-2431. DOI: 10.1002/mrm.29593

Conference Abstracts

2025 **Sherlock, P.**, Duff, M.C., & Brown-Schmidt, S. The impact of TBI on the use of referential context in syntactic ambiguity resolution. *Human Sentence Processing*. College Park, MD, USA. 27-29, March 2025.
[Poster]

- 2024 Volante, C., Diachek, E., **Sherlock, P.**, & Brown-Schmidt, S. Referential modification and memory for object features. *Human Sentence Processing*. Ann Arbor, MI, USA. 16-18, May 2024. [Poster]
- 2022 **Sherlock, P.**, Lynch, K., Choupan, J., & Toga, A. Anterior and middle cerebral artery caliber and PVS size in young adults. *Organization for Human Brain Mapping*. Glasgow, Scotland, UK. 19-23, June 2022. [Poster]

Guest Lectures

- 2026 Collecting Online Behavioral Data
MIT Building Opportunities for Open Sharing and Teaching (BOOST)
Facilitator: Dr. Mehrdad Jazayeri
- 2024 Applying to Postbac Research Positions
Summer Professional Development Series for Undergraduate Trainees
Facilitator: Dr. Whitney Barnett, Vanderbilt University
- 2024 Language and Conversational Memory in Law and Policy
PSY-PC 7500: Advanced Topical Seminar: Cognition in the Real World
Instructor: Dr. Lisa Fazio, Vanderbilt University

Additional Training

- 2022 NASSLLI
Volunteer and attendee, University of Southern California

Outreach

- 2026 “Language in the Mind and Brain” Course
MIT Educational Studies Program HSSP, Cambridge, MA
- 2026 Field Trip Volunteer
MIT Museum, Cambridge, MA
- 2026 Cambridge School Volunteers NetPal
Cambridge Street Upper School, Cambridge, MA
- 2025 “What it Means to be Human” Course
MIT Educational Studies Program Splash, Cambridge, MA

2024 “The Science of Language and Bilingualism” Science Fair Booth
Glendale Spanish Immersion Elementary School, Oak Hill, TN

Awards

2025 McGovern Institute for Brain Research SPOT Award

Technical Proficiency

Coding	Python (PsychoPy, PyLink, Pandas), Bash, Git, SLURM, MATLAB
Methodologies	MRI, Eye-Tracking
Data Collection	jsPsych, Prolific, Qualtrics, REDCap, SONA, Gorilla, PCIBex
Analysis Tools	SPM, CONN, R
Applications	Jupyter, Praat, Microsoft Office Suite, Google Workspaces, Gephi

Natural Languages

English	Native
Spanish	Full working proficiency
Portuguese	Limited working proficiency
Italian	Basic